

UNIT 1: BASIC CONCEPTS

Core Infrastructure, Architectures, and Paradigms of
Information Systems

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1. Introduction to Information Systems Infrastructure

An Information System (IS) is a formal, socio-technical, organizational system designed to collect, process, store, and distribute information. In a contemporary digital ecosystem, information systems form the neural backbone of enterprises, public institutions, and personal utility structures. To fully appreciate how information systems function, one must analyze the foundational components that constitute their infrastructure.

This infrastructure is comprised of several interacting layers. At the lowest abstraction layer lies the physical hardware—the silicon chips, circuit boards, and storage arrays that execute instructions and hold electronic states. Immediately above the hardware is the systems software layer, which manages resource allocation, establishes security boundaries, and exposes clean abstractions to developers and end-users. Above this sits the application software layer, which addresses specific operational, managerial, or computational business needs. Pervading all these components is communications technology, which connects isolated nodes into unified, low-latency fabrics capable of global information distribution.

Core Axiom of Information Systems

The efficacy of an Information System is constrained by its weakest infrastructural layer. High-performance application software cannot mask poor underlying database schemas, nor can optimized code overcome bottlenecked network topologies or obsolete processing components.

Throughout these notes, each component of Unit-1 will be systematically examined, covering both historical contexts and formal technical mechanics. From the historical transition of electronic data processing to modern hyper-scalable cloud paradigms, this document serves as an exhaustive reference text.

2. Hardware Fundamentals and Architectures

Computer hardware comprises the physical, tangible parts of a computer system. The design and arrangement of these components are governed by Computer Architecture, which specifies operational capabilities and programming models. The predominant paradigm for modern computers remains the Von Neumann Architecture, proposed by mathematician and physicist John von Neumann in 1945.

THE VON NEUMANN ARCHITECTURE

The Von Neumann architecture relies on the stored-program concept, where program instructions and data are stored together in the same physical memory space. This model consists of four core building blocks:

- **Central Processing Unit (CPU):** The core computational engine that interprets and executes instructions. It contains the Control Unit (CU), the Arithmetic Logic Unit (ALU), and high-speed registers.
- **Memory Unit:** Main memory (Random Access Memory, or RAM) that holds both data and execution codes during system operation.
- **Input Mechanisms:** Subsystems that convert external physical signals into structured digital representations (e.g., keyboards, scanners, sensors).
- **Output Mechanisms:** Subsystems that translate digital states into human-perceptible or externally actionable forms (e.g., monitors, actuators, printers).

A fundamental physical limitation of this design is the *Von Neumann Bottleneck*. Because instructions and data must travel across the same shared system bus, the CPU frequently sits idle waiting for memory read/write cycles to complete. Modern architectures mitigate this via multi-tier caching architectures (L1, L2, L3 caches) built directly into or near the CPU die using high-speed Static RAM (SRAM).

THE CENTRAL PROCESSING UNIT (CPU) MECHANICS

The CPU executes instructions via a continuous, cyclic process called the **Fetch-Decode-Execute Cycle**:

1. **Fetch:** The Control Unit fetches the next instruction from the memory address stored in the Program Counter (PC) register and places it into the Instruction Register (IR). The PC is then incremented to point to the next instruction.
2. **Decode:** The Control Unit interprets the operational code (opcode) of the instruction to determine what actions are required (e.g., addition, memory load, conditional branch).
3. **Execute:** The ALU performs the requested mathematical or logical operation. If data memory access is needed, memory addresses are referenced. The final output is written to internal registers or back to primary memory.

PRIMARY AND SECONDARY STORAGE TAXONOMY

Storage components within an information system are organized into a strict hierarchy based on speed, capacity, volatility, and cost-per-bit:

Storage Category	Technology Type	Volatility	Typical Access Speed	Primary Use Case
Registers	Flip-Flops within CPU	Volatile	< 1 nanosecond	Immediate arithmetic variables and instruction pointers.
Cache Memory	Static RAM (SRAM)	Volatile	1–10 nanoseconds	Storing frequently accessed instructions and data chunks near the CPU.
Main Memory	Dynamic RAM (DRAM)	Volatile	50–100 nanoseconds	Active operating system structures and running software processes.
Solid-State Storage	NAND Flash Memory	Non-Volatile	25–100 microseconds	High-performance persistent storage (SSDs, NVMe drives).
Magnetic Storage	Platters & Magnetic Heads	Non-Volatile	5–15 milliseconds	Bulk, high-capacity secondary enterprise storage (HDDs).
Optical/Tape Storage	Laser Media / Magnetic Tape	Non-Volatile	Seconds to Minutes	Deep-cold archiving and historical backup preservation.

3. Categories and Generations of Computers

Computer evolution is characterized by leaps in underlying physical component technologies and matching changes in computational scale. Analyzing these shifts requires dividing computer history into specific technological generations and structural categories.

THE FIVE COMPUTING GENERATIONS

Each generation represents a fundamental shift in material science and hardware logic design:

- **First Generation (1940–1956): Vacuum Tubes.** These systems relied on glass vacuum tubes to control electrical charges. They were massive, consuming enormous amounts of electricity and radiating extreme heat that caused frequent hardware breakdowns. Programming was done purely in low-level machine code via physical plugboards or punched cards. Examples include ENIAC and UNIVAC I.
- **Second Generation (1956–1963): Transistors.** The invention of the solid-state transistor at Bell Labs replaced fragile vacuum tubes. Transistors were orders of magnitude smaller, faster, cheaper, and more energy-efficient. This era saw the birth of symbolic assembly languages and early high-level programming languages like FORTRAN and COBOL, along with magnetic core primary memory.

- **Third Generation (1964–1971): Integrated Circuits (ICs).** Breakthroughs in semiconductor fabrication enabled engineers to place hundreds of transistors onto a single silicon wafer. This dramatically reduced system size while increasing speed. Operating systems emerged during this period to manage shared computer resources, replacing primitive batch-processing queues with interactive time-sharing systems.
- **Fourth Generation (1971–Present): Very Large Scale Integration (VLSI) & Microprocessors.** VLSI packed millions (and later billions) of transistors onto a single microchip die, creating the standalone microprocessor (such as the Intel 4004). This technological explosion democratized computing, leading directly to personal computers (PCs), laptops, and handheld mobile devices.
- **Fifth Generation (Present & Beyond): Artificial Intelligence & Quantum Architectures.** This generation focuses on parallel processing architectures, neuromorphic computing, and quantum mechanical systems. It aims to transcend traditional binary logic boundaries through quantum superposition and entanglement, enabling the processing of complex, unstructured data streams in parallel.

CATEGORIES OF CONTEMPORARY COMPUTERS

Modern information systems use different computer categories tailored to specific operational demands:

- **Microcomputers (Personal Computers / Smart Devices):** Small, cost-effective computing systems designed for individual use. Powered by single-chip microprocessors, they range from desktop workstations and notebooks to smartphones and embedded Internet of Things (IoT) controllers.
- **Minicomputers (Mid-Range Servers):** Developed in the 1960s to bridge the gap between massive mainframes and small microcomputers. In modern enterprise IT, this space is occupied by departmental servers and distributed network clusters that handle localized processing workloads.
- **Mainframe Computers:** Massive, highly resilient computing platforms designed for high-volume, centralized data processing. Mainframes are characterized by immense Input/Output (I/O) bandwidth and exceptional reliability, making them ideal for transaction-heavy workloads like core banking, insurance underwriting, and airline reservation systems.
- **Supercomputers:** Highly tuned parallel processing arrays built for computationally intensive, scientific workloads. Instead of processing billions of separate small transactions, supercomputers focus their massive processing power on single, monolithic mathematical challenges, such as cryptographic analysis, global climate modeling, aerodynamic simulations, and molecular modeling. Performance is measured in floating-point operations per second (FLOPS).

4. Communications Technology for Information Systems

Communications technology forms the interconnected fabric of modern information systems, allowing separate computing nodes to exchange data reliably. This layer is governed by formal architectures, reference models, and physical communication media.

THE OSI REFERENCE MODEL

To ensure interoperability between heterogeneous networking equipment, the International Organization for Standardization (ISO) established the Open Systems Interconnection (OSI) model. This conceptual framework divides network communications into seven distinct, sequential layers. Each layer serves the layer above it and relies on the layer beneath it:

1. **Physical Layer (Layer 1):** Manages the raw transmission of unstructured bitstreams over physical media. It defines electrical voltages, timing parameters, pin layouts, and fiber-optic light pulses.
2. **Data Link Layer (Layer 2):** Groups raw bits into structured frames and handles node-to-node data transfer. It manages physical MAC (Media Access Control) addressing, basic error detection, and flow control (e.g., Ethernet, Wi-Fi switching protocols).
3. **Network Layer (Layer 3):** Handles routing across independent networks. It determines the optimal physical path for data packets based on logical network addressing (e.g., Internet Protocol [IPv4/IPv6]).
4. **Transport Layer (Layer 4):** Provides reliable, transparent data transfer between end systems. It manages end-to-end flow control, error recovery, segment multiplexing, and packet reassembly (e.g., Transmission Control Protocol [TCP] and User Datagram Protocol [UDP]).
5. **Session Layer (Layer 5):** Manages, maintains, and synchronizes long-lived communication sessions between applications on separate nodes.
6. **Presentation Layer (Layer 6):** Translates data formats between the network layer and the application layer. It handles abstract data transformations, character encoding conversions (e.g., ASCII to UTF-8), compression, and cryptographic encryption/decryption.
7. **Application Layer (Layer 7):** The interface that interacts directly with end-user software applications. It provides standardized network communication services like web browsing (HTTP/HTTPS), file transfers (FTP), and email routing (SMTP).

NETWORK TOPOLOGIES AND SCOPE CLASSIFICATION

Network infrastructures are classified geographically into several main categories:

- **Local Area Networks (LAN):** High-speed networks spanning a limited geographic area, such as a single office, building, or university campus. They feature low latency and high bandwidth, typically utilizing twisted-pair copper or local fiber-optic lines.

- **Metropolitan Area Networks (MAN):** Networks covering an entire municipality or geographic region, often linking multiple corporate or educational facilities across a city.
- **Wide Area Networks (WAN):** Long-haul networks spanning nations, continents, or the entire globe. WANs typically stitch together multiple smaller LAN networks using leased telecom lines, transoceanic fiber cables, or satellite links. The Internet is the preeminent example of a global public WAN.

The geometric layout of network nodes defines its physical or logical **Topology**:

Topology Type	Structural Layout	Primary Advantage	Primary Disadvantage
Bus	Nodes share a single core backbone cable.	Simple and cheap to deploy for small configurations.	A single break in the main backbone disconnects the entire network.
Star	Every node connects directly to a central hub/switch.	An individual cable failure only impacts that single node.	The central switch is a single point of failure for the entire network.
Ring	Nodes form a closed circular loop.	Predictable data transmission delays without packet collisions.	Adding or removing a single node requires disrupting the whole loop.
Mesh	Redundant connections link nodes to multiple other nodes.	Highly resilient; multiple paths allow data to route around failures.	High deployment cost and complex wiring requirements.

5. Software Systems Taxonomy

Software consists of organized collections of digital instructions that direct computer hardware to execute computational or logic operations. Hardware provides raw execution cycles, while software contains the logical intelligence that drives those cycles. Software is broadly divided into two main categories: Systems Software and Application Software.

Systems Software provides the foundation upon which all other software operates. It acts as an intermediary, shielding application programs from the complex, low-level details of hardware control. It manages internal computer resources, schedules processor cycles, controls memory allocations, and handles low-level storage devices. Without systems software, application developers would need to write custom routines for every specific motherboard chip, disk controller, and display adapter they wanted to support.

Application Software interacts directly with the end-user to perform specific tasks, solve real-world problems, or streamline operational workflows. It runs within the environment established by the systems software, using resources provided through managed interfaces like Application Programming Interfaces (APIs).

6. Systems Software & Operating Systems (OS) Architectures

Systems software is an umbrella term encompassing several critical utility layers:

- **Operating Systems (OS):** The primary resource manager of a computer system.
- **Device Drivers:** Highly specialized hardware-translation modules that allow the OS to communicate with physical peripherals like graphics cards, printers, and network interfaces.
- **Utility Programs:** System maintenance utilities like disk defragmenters, diagnostic modules, file compression tools, and system recovery engines.

THE CORE FUNCTIONS OF AN OPERATING SYSTEM

An operating system serves as a virtual machine abstraction and a resource manager. Its primary responsibilities include:

1. Process Management and CPU Scheduling

A process is an active program in execution. The operating system coordinates process execution using scheduling algorithms to maximize CPU utilization and ensure fair resource sharing. In multi-tasking environments, the OS creates the illusion of simultaneous execution by rapidly switching the CPU between processes—a technique called context switching.

2. Memory Management

The operating system tracks every byte of primary memory (RAM) and dynamically allocates it to active application processes. When a process terminates, the OS reclaims that memory. A critical capability is **Virtual Memory**, which allows systems to execute programs larger than their physical RAM. The OS maps memory addresses to a temporary file on secondary storage (a swap space or page file) and moves pages of data back and forth as needed. If this swapping happens excessively, it can cause a performance issue known as *thrashing*.

3. File System Management

The OS abstracts messy physical storage blocks into neat, logical hierarchies of directories and files. The file system manages file storage, metadata (such as file creation dates and access permissions), and directory routing structures. Common production file systems include NTFS (Windows), ext4 (Linux), and APFS (macOS).

4. Device Management (I/O Control)

The OS handles communications with peripheral devices via their respective drivers. It uses buffering, caching, and spooling (e.g., managing print queues) to smooth out speed differences between fast internal processors and slow external hardware devices.

ARCHITECTURAL CLASSIFICATION OF OPERATING SYSTEMS

Operating systems are optimized for different deployment contexts:

- **Batch Operating Systems:** Early computing operating systems that processed batches of similar jobs sequentially without direct user intervention.
- **Time-Sharing / Multi-user Systems:** Systems that allocate CPU time slices to multiple interactive users simultaneously, giving each user the experience of having a dedicated system.
- **Distributed Operating Systems:** Systems that manage a collection of independent computer nodes, presenting them to applications as if they were a single, unified computer system.
- **Real-Time Operating Systems (RTOS):** Specialized operating systems designed for environments where instruction execution timing must be absolutely predictable. In an RTOS, missing a processing deadline by even a millisecond can cause system failure. These are common in automotive braking units, medical life-support systems, and industrial robotics.

7. Application Software Foundations

Application software sits at the top of the software hierarchy, delivering direct value to end-users. It can be categorized by target audience, deployment scale, and functional scope:

PROPRIETARY VS. OFF-THE-SHELF APPLICATION ARCHETYPES

- **General-Purpose Productivity Applications:** Software built for broad target markets to solve common tasks. Examples include word processors, spreadsheets, web browsers, and media players. These applications are highly versatile and mass-produced, making them cost-effective.
- **Tailor-Made (Custom) Software:** Bespoke solutions built to meet the unique operational requirements of a specific organization. While expensive and time-consuming to build, custom software provides an organizational competitive advantage by aligning precisely with corporate workflows.
- **Enterprise Application Frameworks:** Monolithic, integrated enterprise suites designed to coordinate complex processes across large organizations. This category includes *Enterprise Resource Planning (ERP)* platforms for tracking global supply chains, manufacturing schedules, and accounting records, as well as *Customer Relationship Management (CRM)* systems for managing sales pipelines and client communications.

8. Programming Languages & Paradigm Shift (Procedural vs. OOP)

Programming languages provide the structured syntax used to write the instructions that govern software execution. These languages have evolved through several generations, moving from machine-centric abstractions to human-centric models.

GENERATIONS OF PROGRAMMING LANGUAGES

1. **1GL: Machine Language.** Raw binary sequences (0s and 1s) executed directly by CPU logic gates. Writing code at this level is incredibly difficult and highly error-prone.
2. **2GL: Assembly Language.** Replaces binary opcodes with mnemonics like MOV, ADD, and PUSH. Assembly code is hardware-specific and requires a specialized tool called an *assembler* to translate it into executable machine code.
3. **3GL: High-Level Procedural and Declarative Languages.** Languages like C, Fortran, Pascal, and Java. These use readable, English-like syntax and abstract away hardware specifics like CPU registers and memory addresses. They rely on *compilers* or *interpreters* to convert source code into machine code.
4. **4GL: Domain-Specific Non-Procedural Languages.** Highly specialized languages designed to minimize programming effort for specific tasks. For example, SQL (Structured Query Language) allows developers to define *what* data they want to retrieve without detailing the exact procedural steps needed to read it from disk.
5. **5GL: Knowledge-Based & Constraint-Driven Languages.** Programming environments built around artificial intelligence, constraint logic solvers, and neural net synthesis, where developers specify programmatic constraints and goals rather than hard-coding algorithms.

PROCEDURAL VS. OBJECT-ORIENTED PROGRAMMING (OOP) PARADIGMS

The transition from Procedural Programming to Object-Oriented Programming represents one of the most important shifts in software engineering history. As codebases grew larger, traditional procedural models struggled to manage systemic complexity.

Comparison Attribute	Procedural Programming (e.g., C, Pascal)	Object-Oriented Programming (e.g., Java, C++)
Core Philosophy	Focuses on functions, procedures, and sequential execution steps.	Focuses on "objects" that group data and behavior together.
Data-Function Relationship	Data and functions are separated. Functions act on external data.	Data and internal methods are bound together into cohesive objects.
Structural Approach	Top-Down design; big problems are broken down into smaller procedures.	Bottom-Up design; base objects are built and combined into complex systems.
Access Control / Security	Lacks explicit access modifiers; data can often be modified globally.	Provides robust access modifiers (private, protected, public).
Code Reusability	Limited to function calls; duplication often occurs across modules.	Highly extensible through inheritance and polymorphic interfaces.

THE FOUR CORE PRINCIPLES OF OOP

Object-Oriented Programming controls systemic complexity using four key pillars:

- **Encapsulation:** Bundling data fields and the methods that operate on them into a single structural class, while hiding the internal mechanics from outside code. This establishes clean public interfaces and prevents unintended data modification.
- **Inheritance:** A mechanism that allows a new class (a subclass) to inherit properties and behaviors from an existing class (a superclass). This promotes code reuse and creates natural structural hierarchies.
- **Polymorphism:** The ability for different objects to respond to the same method call in unique ways based on their specific underlying type. This allows systems to treat varied objects interchangeably through shared interfaces.
- **Abstraction:** Hiding complex structural implementation details behind simple interfaces. Users interact with a clean public API without needing to understand the underlying code mechanics.

9. Database Systems (DBMS) and Architecture

Data is a critical asset for modern enterprises. Traditional file systems store data in isolated, unstructured flat files, which leads to significant issues like data redundancy (the same data

stored in multiple places), inconsistency (conflicting updates), and tight coupling between data storage formats and application code.

A **Database Management System (DBMS)** solves these issues by providing a centralized software interface to create, retrieve, update, and manage structured data safely and efficiently.

THE THREE-SCHEMA DATABASE ARCHITECTURE

To isolate user applications from physical storage details, the ANSI-SPARC framework defines a clear three-tier database architecture:

1. **External Schema (View Level):** The top layer, describing how individual end-users and applications view the data. A database can support many external views, shielding users from seeing data fields they don't need or aren't authorized to access.
2. **Conceptual Schema (Logical Level):** The middle layer, describing the entire database structure, including all data entities, their attributes, relationships, and security constraints. It models the data logically without considering physical storage limitations.
3. **Internal Schema (Physical Level):** The lowest layer, defining how data is actually stored on persistent disks, including file allocations, indexing tracks, page structures, and compression algorithms.

This separation enables **Data Independence**. The physical storage layout can be changed (e.g., switching to faster storage hardware) without modifying the conceptual schema or forcing rewrites of user application code.

RELATIONAL VS. NON-RELATIONAL DATABASE MODELS

Modern information systems typically use one of two primary database paradigms:

- **Relational Database Management Systems (RDBMS):** Based on relational algebra, these systems organize data into structured tables consisting of rows (tuples) and columns (attributes). Tables are linked together using Primary Keys and Foreign Keys. RDBMS platforms enforce strict **ACID** guarantees:
 - *Atomicity:* Transactions must succeed completely or fail completely; there are no partial saves.
 - *Consistency:* Every transaction must move the database from one valid state to another, respecting all constraints.
 - *Isolation:* Concurrent transactions must execute without interfering with one another.
 - *Durability:* Once a transaction is committed, its changes are permanently saved, even during a system crash.

Examples include PostgreSQL, MySQL, and Oracle Database.

- **Non-Relational (NoSQL) Systems:** Designed to trade strict ACID guarantees for horizontal scalability and flexible schemas, making them ideal for massive volumes of unstructured or semi-structured data. They include *Document Stores* (e.g., MongoDB), *Key-Value Frameworks* (e.g., Redis), *Graph Databases* (e.g., Neo4j), and *Wide-Column Stores* (e.g., Cassandra).

10. Electronic Data Processing (EDP) Development Cycles

Electronic Data Processing (EDP) refers to the automated collection, manipulation, and summary of data using electronic computers. Historically, EDP automated manual paper-based accounting and filing practices. Today, it has evolved into high-speed pipeline processing systems that ingest massive, real-time data streams.

THE FOUR STAGES OF THE EDP CYCLE

An EDP pipeline processes data through four fundamental, continuous stages:

1. **Data Ingestion / Input:** Gathering raw data from source points (e.g., point-of-sale terminals, IoT sensors, manual web forms) and converting it into a clean digital format. This stage often includes data validation routines to catch formatting errors early.
2. **Data Processing:** Manipulating and transforming the raw input data using systematic business rules and mathematical operations. Processing typically happens via two main models:
 - *Batch Processing:* Accumulating large volumes of transaction records over time and processing them all at once during scheduled, low-demand hours (e.g., end-of-day bank clearing).
 - *Real-Time Stream Processing:* Processing transactions immediately as they arrive, providing sub-second analytical feedback (e.g., fraud detection systems checking credit card activity).
3. **Storage and Retrieval:** Writing processed records systematically into databases or enterprise data warehouses, ensuring data remains accessible for future queries while protecting it against unauthorized modification.
4. **Data Output / Distribution:** Converting processed information into actionable forms for end-users, such as interactive management dashboards, automated email alerts, billing statements, or structural analytical reports.

11. Open Source vs. Closed Source Economics & Architectures

Software distribution models influence how software is built, licensed, and deployed across enterprise information systems.

Evaluation Dimension	Open Source Software (OSS)	Closed Source (Proprietary) Software
Source Code Accessibility	Publicly available; anyone can audit, modify, and recompile the code.	Strictly protected corporate secret; only accessible to the original developer.
Licensing & Acquisition	Governed by flexible licenses (e.g., GNU GPL, MIT, Apache) with no upfront purchase fees.	Requires commercial licenses, often tied to per-user, per-core, or subscription models.
Innovation Cycle	Driven by global community contributions and rapid open-peer code reviews.	Driven by internal corporate product maps and commercial market strategies.
Support Infrastructure	Relies on community forums, open documentation, and third-party enterprise service agreements.	Provides structured corporate support via SLAs and dedicated helpdesks.
Vendor Lock-in Risk	Minimal; organizations can fork code or self-host if needed.	High; migrating away can require costly software rewrites.

12. Cloud Computing Architecture and Distributed Paradigms

Cloud Computing has shifted information systems from localized physical hardware to on-demand utility computing models. The National Institute of Standards and Technology (NIST) defines cloud computing through five essential characteristics, three service models, and four deployment models.

THE FIVE ESSENTIAL CHARACTERISTICS OF CLOUD COMPUTING

- **On-Demand Self-Service:** Users can provision computing resources (such as server time or network storage) automatically as needed, without requiring human interaction with the service provider.
- **Broad Network Access:** Capabilities are available over the network and accessed through standard mechanisms from diverse client platforms (e.g., mobile phones, laptops, and workstations).
- **Resource Pooling:** The provider's physical computing resources are pooled together to serve multiple consumers using a multi-tenant model. Physical and virtual resources are dynamically assigned and reassigned according to consumer demand.
- **Rapid Elasticity:** Resources can be provisioned and released elastically—sometimes automatically—to scale rapidly outward or inward to match real-time demand variations.

- **Measured Service:** Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts).

CLOUD SERVICE MODELS (THE SPI FRAMEWORK)

Cloud infrastructure services are organized into three primary layers based on user control:

- **Infrastructure as a Service (IaaS):** Provides fundamental computing resources over the web, including physical or virtual machines, raw storage volumes, virtual local networks, and firewalls. The consumer manages the operating system, middleware, and applications, while the provider maintains the physical datacenters and virtualization layers. Examples: AWS EC2, Google Compute Engine, Microsoft Azure VMs.
- **Platform as a Service (PaaS):** Delivers a managed development environment. The provider manages the operating system, database engines, and network routing, allowing the customer to focus entirely on writing and deploying application code. Examples: Heroku, AWS Elastic Beanstalk, Google App Engine.
- **Software as a Service (SaaS):** Delivers fully functional applications over the web through a browser or client interface. The end-user has minimal control over application configurations or underlying infrastructure. Examples: Google Workspace, Microsoft 365, Salesforce.

CLOUD DEPLOYMENT MODELS

- **Public Cloud:** Infrastructure is owned by a third-party cloud provider and shared across multiple tenants over the public internet. Offers high scalability and low upfront capital expenditure.
- **Private Cloud:** Infrastructure is provisioned for exclusive use by a single organization. It can be owned, managed, and operated by the organization itself or a third party, and may exist on-premise or off-premise, providing maximum security and data control.
- **Hybrid Cloud:** Combines public and private cloud environments, allowing data and applications to be shared between them. This enables organizations to run core secure workloads on private infrastructure while bursting to the public cloud to handle traffic spikes.
- **Multi-Cloud:** The strategic use of cloud services from multiple independent cloud providers (e.g., using AWS for computation, Azure for identity verification, and Google Cloud for analytics) to prevent vendor lock-in and maximize system resilience.